Course Name: M. Math, 1^{st} year
Subject Name : Topology II
Assignment 3

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1. Let G be a topological group with multiplication $m : G \times G \to G$. Let ω denote the composite
$$G \vee G \subset G \times G \xrightarrow{m} G.$$
Calculate the effect of ω on homology groups using the isomorphism $\tilde{H}_*(G \vee G) \cong \tilde{H}_*(G) \oplus \tilde{H}_*(G)$.

2. Let $\Delta : S^n \to S^n \times S^n$ be the diagonal map given by $\Delta(x) = (x, x)$. Calculate $H_*(\text{Cone}(\Delta))$, the homology of the mapping cone of Δ.

3. Let f be a map $S^n \to S^k$, $n > k$.
a) Calculate $H_*(C(f))$ where $C(f)$ is the mapping cone of f.
b) Calculate $H_*(M(f))$ where $M(f)$ is the mapping cylinder of f.
c) Consider the quotient map q from $M(f)$ to $C(f)$. Calculate the effect of q_* on homology groups.

4. Find the homology $H_*(S^n, \{p_1, \cdots, p_k\})$ where $p_1, \cdots, p_k$ are k distinct points on S^n.

5. a) For a point p in S^n find $H_*(S^n, S^n - p)$.
b) Do the same exercise for the Torus, and complex projective space.

6. Let T be the torus which is a CW complex whose 1-skeleton is $S^1 \vee S^1$ (denoted by a and b). Compute $H_* T/S^1_a$.

7. Show that the map ϕ_k $z \mapsto z^k$ from S^1 to S^1 induces multiplication by k on H_1.

8. Let $\phi_{n,k}$ denote $\Sigma^{n-1}\phi_k : S^n \to S^n$. Compute $H_*(C(\phi_{n,k})$, where $C(\phi)$ is the mapping cone of ϕ.

9. Prove that the sphere S^n does not retract to it's equator.

10. Let X be a sΔ-set such that $X_k = \varnothing$ if $k > n$. Show that $H_n(|X|)$ is a free Abelian group.

11. Find a space X with $H_0(X) = \mathbb{Z}$, $H_1(X) = \mathbb{Z}$, $H_2(X) = \mathbb{Z}/2$ and $H_i(X) = 0$ if $i > 2$.

12. Find the homology groups of the Klein bottle.

13. Let X be a space and p be the cone point of $C(X)$. Compute $H_*(C(X), C(X) - p)$.

14. Let C be the circle on the torus T which is the image of the line $px = qy$ under the covering map $\mathbb{R}^2 \to T$ [this identifies $T = S^1 \times S^1$ and sends (a, b) to their image modulo $\mathbb{Z} \times \mathbb{Z}$]. Let $X = T/C$ the quotient space obtained by identifying C to a point. Calculate $H_*(X)$.

15. Calculate $H_*(S^3 \times S^5)$, $H_*(S^3 \vee S^5)$, and the map induced by the inclusion $S^3 \vee S^5 \to S^3 \times S^5$.

16. Calculate $H_*(M(n))$ where $M(n)$ is obtained by attaching a 2-cell to S^1 via a map of degree n.

17. a) Suppose X is an n-dimensional CW complex with exactly one n-cell. We say that the top cell e^n splits off from X if the attaching map $\partial e^n \to X^{(n-1)}$ is null homotopic. Show that in this case $X \simeq X^{(n-1)} \vee S^n$.
b) Consider the CW complex structure on $\mathbb{R}P^{2n}$ with one i-cell for every $0 \leq i \leq 2n$. Show that the top cell does not split off from $\mathbb{R}P^{2n}$.

18. Let $M(\mathbb{Z}/n, k) = C(f_n : S^k \to S^k)$ where f_n is a map of degree n.
a) Describe a CW complex structure on $M(\mathbb{Z}/2, 5) \times M(\mathbb{Z}/3, 7)$.
b) Compute $H_*(M(\mathbb{Z}/2, 5) \times M(\mathbb{Z}/3, 7))$.

19. Write down a map f from the torus T to itself such that the map $f_* : H_1(T) \to H_1(T)$ is given by the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

20. Consider the map $\phi : \mathbb{R}P^2 \to \mathbb{R}P^2$ given by $\phi([x_0, x_1, x_2]) = [x_0^3, x_1^3, x_2^3]$. Compute the action of ϕ_* on homology groups.

21. Let $M(\mathbb{Z}/n,k) = C(f_n)$ where $f_n : S^k \to S^k$ is a map of degree n. ($C(f)$ denotes the cone of f)
a) Compute $H_*(M(\mathbb{Z}/2,5);\mathbb{Z})$ and $H_*(M(\mathbb{Z}/2,5);\mathbb{Z}/2)$.
b) Consider the composite $S^5 \xrightarrow{f} S^5 \xrightarrow{g} S^5$ where f, g are maps of degree 2. Show that we obtain a map ψ as the composite
$$M(\mathbb{Z}/4,5) \simeq C(g \circ f) \to C(g) \simeq M(\mathbb{Z}/2,5)$$
(i.e, verify the existence of the arrows and the homotopy equivalences)
d) Compute the effect of ψ_* on homology groups.

22. Compute $H_*(\mathbb{R}P^n, \mathbb{R}P^n - \mathbb{R}P^i; \mathbb{Z}/2)$.

23. Let p, q be two points in S^2, $p \neq q$. Suppose $r \in S^2$ is different from p and q. Find $\pi_1(S^2 - p - q, r)$.

24. Let p, q be two points in the torus T, $p \neq q$. Suppose $r \in T$ is different from p and q. Find $\pi_1(T - p - q, r)$.

25. Calculate $H_*(X \times \Sigma_g)$ if X is a CW complex.

26. Recall that $M(\mathbb{Z}/k,n) = \mathrm{Cone}(f : S^n \to S^n)$ where degree of f is k.
a) Prove that $M(\mathbb{Z}/k,n) \vee M(\mathbb{Z}/l,m) \to M(\mathbb{Z}/k,n) \times M(\mathbb{Z}/l,m)$ induces isomorphism in H_* if $(k,l) = 1$.
b) What happens if $(k,l) \neq 1$?

27. Let X be a simply connected space. Prove that any map from X to $(S^1)^k$ induces a 0 map on reduced homology.

28. Prove that any map $T \to T$ (T = Torus) whose restriction to $S^1 \vee S^1$ is null-homotopic induces a 0 map on reduced homology.

29. Compute the homology of $\mathbb{R}P^{n+k}/\mathbb{R}P^n$.

30. Consider the exact sequence $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$. Prove that there is a long exact sequence
$$\cdots \to H_k(X) \xrightarrow{2} H_k(X) \to H_k(X;\mathbb{Z}/2) \to H_{k-1}(X) \to \cdots$$
Compute this sequence for $X = \mathbb{R}P^n$.

31. a) Consider $TS^2 = \{(x,v) | x \in S^2, v \in \mathbb{R}^3, \langle v, x\rangle = 0\} \subset S^2 \times \mathbb{R}^3$. Show that TS^2 deformation retracts to $A = \{(x,0) | x \in S^2\} \subset TS^2$, which is homeomorphic to S^2.
b) Let $X = \{(x,v) \in TS^2 | |v| = 1\}$. Show that $TS^2 - A$ deformation retracts to X.
c) Consider $\pi_1 : TS^2 \to S^2$ the projection onto the first factor. Show that any $s : S^2 \to TS^2$ such that $\pi_1 \circ s = id$, satisfies $Im(s) \cap A \neq \phi$.

32. Let $q : S^n \to \mathbb{R}P^n$ be the usual quotient map.
a) Compute $q_* : H_n(S^n) \to H_n(\mathbb{R}P^n)$.
b) Compute $q_* : H_n(S^n;\mathbb{Z}/2) \to H_n(\mathbb{R}P^n;\mathbb{Z}/2)$.

33. Let X be a path connected space such that $\pi_1(X)$ is a finite group. Prove that for any map $X \to S^1 \times S^1$, the induced map $f_* : \tilde{H}_*(X) \to \tilde{H}_*(S^1 \times S^1)$ is 0.

34. a) Prove that every map $\mathbb{R}P^{2n} \to \mathbb{R}P^{2n}$ has a fixed point.
b) Construct maps $\mathbb{R}P^{2n-1} \to \mathbb{R}P^{2n-1}$ without fixed points.

35. Write the group C_2 as $\{1, \sigma\}$ such that $\sigma^2 = 1$. Consider the C_2 action on $\mathbb{C}P^2 \times S^2$ given by $\sigma([z_0, z_1, z_2], x) = ([\bar{z_0}, \bar{z_1}, \bar{z_2}], -x)$. Define P as the space $\mathbb{C}P^2 \times S^2/C_2$.
a) Show that P is path-connected.
b) Prove that the quotient map $\mathbb{C}P^2 \times S^2 \to P$ is a covering space.
c) Compute $\pi_1(P, p)$ for any point p.

36. Let C_2 (the cyclic group of order 2 generated by σ) act on $S^n \times S^n$ according to the formula $\sigma(x, y) = (-x, -y)$. Let $P = (S^n \times S^n)/C_2$.
a) Prove that the quotient map $S^n \times S^n \to P$ is a covering space.
b) Write down a map $q : P \to \mathbb{R}P^n \times \mathbb{R}P^n$ which is a covering space.
c) Prove that every map from P to S^1 is homotopic to a constant map.

37. Let $f_k : S^n \to S^n$ be a map of degree k. Consider $q \circ f_k : S^n \to \mathbb{R}P^n$. Compute the homology $H_*(Cone(q \circ f_k))$.

38. Let $f_k : S^{2n+1} \to S^{2n+1}$ be a map of degree k. Consider $q \circ f_k : S^{2n+1} \to \mathbb{C}P^n$. Compute the homology $H_*(Cone(q \circ f_k))$.

39. Let $\mathbb{C}P^n \to \mathbb{C}P^{n+k}$ be the standard inclusion. Compute the homology of $\mathbb{C}P^{n+k}/\mathbb{C}P^n$.

40. Compute the homology of $\mathbb{C}P^n \times \mathbb{C}P^m$ and $\mathbb{C}P^n \times \mathbb{R}P^m$.

41. Let Γ be a connected graph with a base point. For a based X compute $H_*(X \wedge \Gamma)$.

42. Let Y be a finite based CW complex with cells in only even degrees.
a) Show that $H_i Y \cong \mathbb{Z}^{r_i}$ where r_i is the number of i-cells.
b) For any space X compute $H_*(X \times Y)$ and $\tilde{H}_*(X \wedge Y)$.